SQL SERVER | DATA ANALYTICS PROJECT

# Data Cleaning in SQL

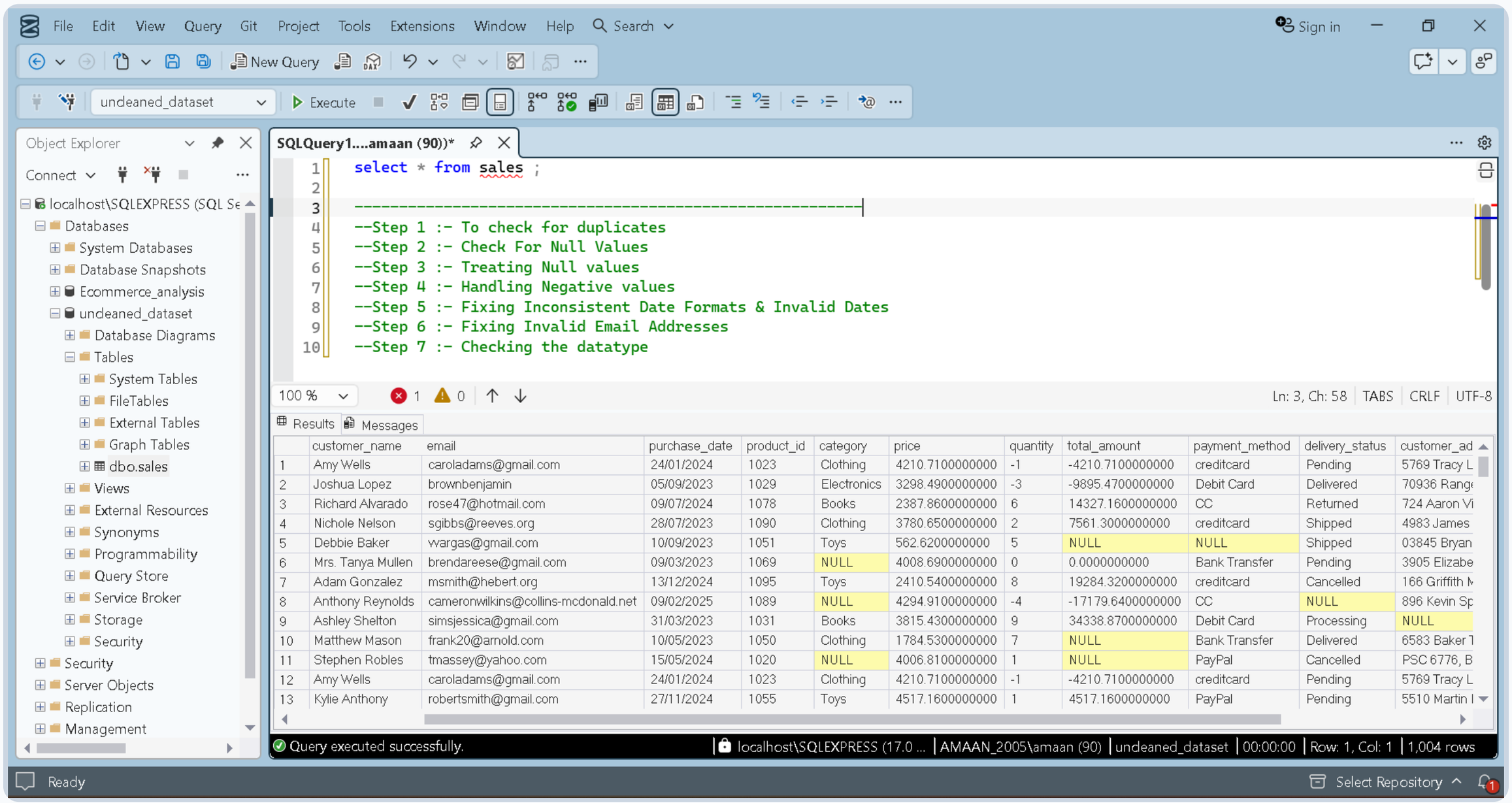
Real SQL Server (SSMS) screenshots,
7 steps, 1,004 messy rows to 1,000 clean

**By Amaan**

Swipe > step-by-step walkthrough

# The Raw Data

## Table: sales | 1,004 rows | 13 columns




*Raw data in SSMS: NULLs (yellow), negative quantity, mixed payment labels*

## Problems visible at first look:

- Negative **quantity** and negative **total_amount**
- **NULL** in category, price, total_amount, payment_method...
- Payment labels: **creditcard, CC** instead of Credit Card
- Duplicate rows (Amy Wells appears twice)

# 1 Remove Duplicates

## ROW_NUMBER() inside a CTE

First I found rows where **row_num > 1**. These are repeated transaction_ids:

SQLQuery2....amaan (84))* | SQLQuery1.s...\amaan (90))*

```
1  select * from sales ;
2
3  --Step 1 :- To check for duplicates
4  with CTE as (
5  select * ,
6  ROW_NUMBER() over(partition by transaction_id order by transaction_id) as row_num
7  from sales
8  )
9  select * from CTE
10 WHERE row_num > 1 ;
11
```

100 % | No issues found | Ln: 11, Ch: 1 | TABS | CRLF | UTF-8

Results | Messages

| | r_id | customer_name | email | purchase_date | product_id | category | price | quantity | total_amount | payment_method | delivery_status | customer_address | row_num |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | Amy Wells | caroladams@gmail.com | 24/01/2024 | 1023 | Clothing | 4210.7100000000 | -1 | -4210.7100000000 | creditcard | Pending | 5769 Tracy Lane Port Amy, AZ 35865 | 2 |
| 2 | | Nichole Nelson | sgibbs@reeves.org | 28/07/2023 | 1090 | Clothing | 3780.6500000000 | 2 | 7561.3000000000 | creditcard | Shipped | 4983 James Plains Roachland, NJ 41398 | 2 |
| 3 | | Sierra Lopez | tammydaniels | 26/03/2023 | 1060 | NULL | 1455.3900000000 | -1 | -1455.3900000000 | CC | Processing | 960 Dawn Spurs East Katherine, WY 68562 | 2 |
| 4 | | Todd Jones | hwebb@gmail.com | 14/05/2023 | 1055 | Home & Kitchen | NULL | 7 | NULL | Bank Transfer | Shipped | 2778 Sanchez Port Lake Davidhaven, ID 50126 | 2 |

*4 duplicate rows found*

Then I changed SELECT to **DELETE** on the same CTE:

SQLQuery2....amaan (84))* | SQLQuery1.s...\amaan (90))*

```
from sales
)
select * from CTE
WHERE row_num > 1 ;

-- DUPLETE DUPLICATE DATA
with CTE as (
select * ,
ROW_NUMBER() over(partition by transaction_id order by transaction_id) as row_num
from sales
)
delete from CTE
where row_num > 1 ;
```

100 % | No issues found

Messages

```
(4 rows affected)

Completion time: 2026-10-25T12:07:48.1574783+05:30
```

*(4 rows affected) | 1,004 rows become 1,000*

> **Analyst tip:** always SELECT first, check the rows, and only then DELETE.

# 2 Check for NULL Values

Dynamic SQL, one query for all columns

Instead of writing IS NULL for every column, I built the query from **INFORMATION_SCHEMA.COLUMNS** with STRING_AGG and ran it with **sp_executesql**.

SQLQuery16...maan (77))* | SQLQuery15....\amaan (91))* | REMOVE NULL...amaan (90))* | Analysis.sql ...5\amaan (88))*

```sql
--Step 2 :- Check For Null Values

DECLARE @SQL NVARCHAR(MAX) = '';

SELECT @SQL = STRING_AGG(
    'SELECT ''' + COLUMN_NAME + ''' AS ColumnName,
    COUNT(*) AS NullCount
    FROM ' + QUOTENAME(TABLE_SCHEMA) + '.sales
    WHERE ' + QUOTENAME(COLUMN_NAME) + ' IS NULL',
    ' UNION ALL '
)
WITHIN GROUP (ORDER BY COLUMN_NAME)
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'sales';

-- Execute the dynamic SQL
EXEC sp_executesql @SQL;
```

80 % | No issues found | Ln: 3, Ch: 1 | (422 chars, 15 lines) | SPC | CRLF | UTF-8

Results | Messages

| | ColumnName | NullCount |
|---|---|---|
| 1 | category | 0 |
| 2 | customer_address | 0 |
| 3 | customer_id | 3 |
| 4 | customer_name | 0 |
| 5 | delivery_status | 0 |
| 6 | email | 102 |
| 7 | payment_method | 0 |
| 8 | price | 0 |
| 9 | product_id | 0 |
| 10 | purchase_date | 0 |
| 11 | quantity | 0 |
| 12 | total_amount | 0 |

Query executed successfully. | localhost\SQLEXPRESS (17.0 ... | AMAAN_2005\amaan (77) | uncleaned_dataset | 00:00:00 | Row: 1, Col: 1 | 13 rows

*NullCount for each column*

> **Why it matters:** run this before and after cleaning to prove your fixes worked.

3

# Treat NULL Values (Text)

## Category, address, payment, delivery status

Treating Nu...amaan (92))* | SQLQuery11.s...\amaan (70)) | SQLQuery9.s...ot connected* | SQLQuery8.s...ot connected*

```sql
--Step 3 :- Treating Null values

SELECT DISTINCT category from sales

-----------------category
UPDATE sales
SET category='Unknown'
WHERE category IS NULL

-----------------customer_address
UPDATE sales
SET customer_address='Not Available'
WHERE customer_address IS NULL

-----------------payment_method

SELECT DISTINCT payment_method from sales

UPDATE sales
SET payment_method='Credit Card'
WHERE payment_method IN ('creditcard','CC','credit')

UPDATE sales
SET payment_method='Cash'
WHERE payment_method IS NULL

-----------------delivery_status

SELECT DISTINCT delivery_status from sales

UPDATE sales
SET delivery_status='Not Delivered'
WHERE delivery_status IS NULL
```

80 % | No issues found | Ln: 27, Ch: 33 | TABS | CRLF | UTF-8

Results | Messages

Query executed successfully. | localhost\SQLEXPRESS (17.0 ... | AMAAN_2005\amaan (92) | uncleaned_dataset | 00:00:00 | Row: 1, Col: 1 | 1,973 rows

*Fill with clear labels and standardise payment_method*

Treating Nu...amaan (55))* | CHECK NULL ...\amaan (85))* | CHI

```sql
-----------------delivery_status

SELECT DISTINCT delivery_status from sales

UPDATE sales
SET delivery_status='Not Delivered'
WHERE delivery_status IS NULL
```

100 % | No issues found

Results | Messages

| | delivery_status |
|---|---|
| 1 | Pending |
| 2 | NULL |
| 3 | Returned |
| 4 | Delivered |
| 5 | Processing |
| 6 | Cancelled |
| 7 | Shipped |

*delivery_status: distinct values include NULL, then UPDATE to Not Delivered*

> **Assumption:** missing payment_method set to Cash. Always document such assumptions.

3

# Treat NULL Values (Price)

## Mean, median, mode, then category-wise average

Analysis.sq...\amaan (88))* | Treating Null...5\amaan (92))* | SQLQuery11.s...\amaan (70)) | SQLQuery9.s...ot connected*

```
1  ----MEAN
2
3  SELECT AVG(price) from sales
4  ---2510.76
5
6  ----MODE
7
8  SELECT price,count(*) as max_count
9  from sales
10 group by price
11 order by max_count desc
12
13 ----Median
14 --2530.75
15
16 SELECT DISTINCT
17   PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY price) OVER() AS median
18 FROM sales;
19
20 ----------------------------------------
21 select category, avg(price) as avg_price
22 from sales
23 group by category
24
25 --Unknown            2511.416405
26 --Books              2574.457346
27 --Home & Kitchen     2507.058378
28 --Toys               2235.471689
29 --Electronics        2663.927840
30 --Clothing           2539.278187
31
```

80 % | No issues found | Ln: 31, Ch: 1 | TABS | CRLF | UTF-8

Connected. (1/1) | localhost\SQLEXPRESS (17.0 ... | AMAAN_2005\amaan (88) | uncleaned_dataset | 00:00:00 | Row: 0, Col: 0 | 0 rows

*Mean 2510.76 | Median 2530.75 | average price per category*

REMOVE NU...maan (90))* | Analysis.sql ...5\amaan (88))* | Treating Null...5\amaan (92))* | SQLQuery11.s...\amaan (70))

```
1  --Unknown
2  UPDATE sales
3  SET price=2511.41
4  WHERE price is NULL and category='Unknown'
5
6  --Books
7  UPDATE sales
8  SET price=2574.45
9  WHERE price is NULL and category='Books'
10
11 --Home & Kitchen     2507.05
12 UPDATE sales
13 SET price=2507.05
14 WHERE price is NULL and category='Home & Kitchen'
15
16 --Toys               2235.47
17 UPDATE sales
18 SET price=2235.47
19 WHERE price is NULL and category='Toys'
20
21 --Electronics        2663.92
22 UPDATE sales
23 SET price=2663.92
24 WHERE price is NULL and category='Electronics'
25
26 ---Clothing          2539.27
27 UPDATE sales
28 SET price=2539.27
29 WHERE price is NULL and category='Clothing'
30
31 select * from sales;
```

85 % | No issues found | Ln: 31, Ch: 21 | TABS | CRLF | UTF-8

Connected. (1/1) | localhost\SQLEXPRESS (17.0 ... | AMAAN_2005\amaan (90) | uncleaned_dataset | 00:00:00 | Row: 0, Col: 0 | 0 rows

*Each missing price filled with its own category average*

**Analyst tip:** group-wise fill is more accurate than one global average.

4

# Handle Negative Values

## Find them, then convert with ABS()

HANDLE NE...maan (52))* | Treating Null...5\amaan (69))*

```
1  --Step 4 :- Handling Negative values
2  select quantity from sales
3  where quantity < 0;
4
```

100 % — No issues found

Results | Messages

| | quantity |
|---|---|
| 1 | -1 |
| 2 | -3 |
| 3 | -4 |
| 4 | -1 |
| 5 | -1 |
| 6 | -5 |
| 7 | -2 |
| 8 | -4 |
| 9 | -2 |
| 10 | -4 |
| 11 | -2 |
| 12 | -4 |
| 13 | -1 |
| 14 | -3 |
| 15 | -4 |
| 16 | -1 |
| 17 | -1 |
| 18 | -5 |
| 19 | -5 |
| 20 | -1 |
| 21 | -3 |

Query executed successfully.

*SELECT quantity WHERE quantity < 0*

HANDLE NE...maan (52))* | Treating Null...5\amaan (69))* | CHECK NULL ...ot connected* | CHECK NULL ...ot connected*

```
1  --Step 4 :- Handling Negative values
2  select * from sales
3  where quantity < 0;
4
5  update sales
6  set quantity = abs(quantity )
7  where quantity < 0;
```

100 % — No issues found — Ln: 2, Ch: 21 | TABS | CRLF | UTF-8

Results | Messages

| | transaction_id | customer_id | customer_name | email | purchase_date | product_id | category | price | quantity | total_amount | payment_metho |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1001 | 102 | Amy Wells | caroladams@gmail.com | 24/01/2024 | 1023 | Clothing | 4210.7100000000 | 1 | -4210.7100000000 | Credit Card |
| 2 | 1002 | 435 | Joshua Lopez | brownbenjamin | 05/09/2023 | 1029 | Electronics | 3298.4900000000 | 3 | -9895.4700000000 | Debit Card |
| 3 | 1003 | 348 | Richard Alvarado | rose47@hotmail.com | 09/07/2024 | 1078 | Books | 2387.8600000000 | 6 | 14327.1600000000 | Credit Card |
| 4 | 1004 | 270 | Nichole Nelson | sgibbs@reeves.org | 28/07/2023 | 1090 | Clothing | 3780.6500000000 | 2 | 7561.3000000000 | Credit Card |
| 5 | 1005 | 106 | Debbie Baker | vvargas@gmail.com | 10/09/2023 | 1051 | Toys | 562.6200000000 | 5 | NULL | Credit Card |
| 6 | 1006 | 71 | Mrs. Tanya Mullen | brendareese@gmail.com | 09/03/2023 | 1069 | UNKNOWN | 4008.6900000000 | 0 | 0.0000000000 | Bank Transfer |
| 7 | 1007 | 188 | Adam Gonzalez | msmith@hebert.org | 13/12/2024 | 1095 | Toys | 2410.5400000000 | 8 | 19284.3200000000 | Credit Card |
| 8 | 1008 | 20 | Anthony Reynolds | cameronwilkins@collins-mcdonald.net | 09/02/2025 | 1089 | UNKNOWN | 4294.9100000000 | 4 | -17179.6400000000 | Credit Card |
| 9 | 1009 | 102 | Ashley Shelton | simsjessica@gmail.com | 31/03/2023 | 1031 | Books | 3815.4300000000 | 9 | 34338.8700000000 | Debit Card |
| 10 | 1010 | 121 | Matthew Mason | frank20@arnold.com | 10/05/2023 | 1050 | Clothing | 1784.5300000000 | 7 | NULL | Bank Transfer |
| 11 | 1011 | 466 | Stephen Robles | tmassey@yahoo.com | 15/05/2024 | 1020 | UNKNOWN | 4006.8100000000 | 1 | NULL | PayPal |
| 12 | 1012 | 214 | Kylie Anthony | robertsmith@gmail.com | 27/11/2024 | 1055 | Toys | 4517.1600000000 | 1 | 4517.1600000000 | PayPal |

*UPDATE quantity = ABS(quantity)*

**Think like an analyst:** a negative quantity can also mean a return. Confirm with the business.

# 4 Fix Totals & Customer Names

Recalculate total_amount = price x quantity

SQLQuery15...maan (91))* | REMOVE NULL...amaan (90))* | Analysis.sql ...5\amaan (88))* | Treating Null...5\amaan (92))*

```sql
--Step 4 :- Handling Negative values

SELECT * FROM SALES
where quantity <0

UPDATE sales
SET quantity = ABS(quantity)
WHERE quantity <0

UPDATE sales
SET total_amount= price*quantity
WHERE total_amount IS NULL OR total_amount <> price*quantity



SELECT * FROM SALES
WHERE customer_name iS NULL

update sales
set customer_name='User'
where customer_name is NULL

select * from sales
where customer_name = 'user'
```

80 % | No issues found | Ln: 21, Ch: 28 | TABS | CRLF | UTF-8

Results | Messages

| | transaction_id | customer_id | customer_name | email | purchase_date | product_id | category | price | quantity | total_amount | payment_method | del |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1039 | 48 | user | johnsontamara@maddox.info | 13/08/2023 | 1035 | Books | 4665.7200000000 | 5 | 23328.6000000000 | Credit Card | Nc |
| 2 | 1045 | 463 | user | alanmurphy@hotmail.com | 31/12/2023 | 1030 | Electronics | 1466.1100000000 | 2 | 2932.2200000000 | PayPal | De |
| 3 | 1081 | 34 | user | thomasanita@yahoo.com | 27/02/2024 | 1028 | Clothing | 2539.2700000000 | 6 | 15235.6200000000 | Credit Card | Nc |
| 4 | 1092 | 259 | user | stevenkelly@hotmail.com | 2024-02-30 | 1056 | Home & Kitchen | 2253.0100000000 | 4 | 9012.0400000000 | Debit Card | Pr |
| 5 | 1110 | 251 | user | NULL | 2024-02-30 | 1003 | Clothing | 3272.8800000000 | 0 | 0.0000000000 | Bank Transfer | Nc |

Query executed successfully. | localhost\SQLEXPRESS (17.0 ... | AMAAN_2005\amaan (91) | uncleaned_dataset | 00:00:00 | Row: 1, Col: 1 | 53 rows

*Update total_amount, then fill missing customer_name*

- Price and quantity are fixed **first**, then total_amount is rebuilt.
- Missing names set to **User**. 53 rows in the raw file.

# 5 Fix Dates

## Invalid dates like 2024-02-30

```sql
--Step 5 :- Fixing Inconsistent Date Formats & Invalid Dates

select * from sales
WHERE purchase_date ='2024-02-30'

update sales
set purchase_date =
  case when TRY_CONVERT(date,purchase_date,103 ) is not null
  then TRY_CONVERT(date,purchase_date,103)
  else null
  end;

  use uncleaned_dataset;
  select * from sales;
```

100 % | No issues found | Ln: 14, Ch: 1 | (22 chars, 1 lines) | TABS | CRLF | UTF-8

Results | Messages

| | transaction_id | customer_id | customer_name | email | purchase_date | product_id | category | price | quantity | total_amount | payment_methoc |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1001 | 102 | Amy Wells | caroladams@gmail.com | 24/01/2024 | 1023 | Clothing | 4210.7100000000 | 1 | 4210.7100000000 | Credit Card |
| 2 | 1002 | 435 | Joshua Lopez | NULL | 05/09/2023 | 1029 | Electronics | 3298.4900000000 | 3 | 9895.4700000000 | Debit Card |
| 3 | 1003 | 348 | Richard Alvarado | rose47@hotmail.com | 09/07/2024 | 1078 | Books | 2387.8600000000 | 6 | 14327.1600000000 | Credit Card |
| 4 | 1004 | 270 | Nichole Nelson | sgibbs@reeves.org | 28/07/2023 | 1090 | Clothing | 3780.6500000000 | 2 | 7561.3000000000 | Credit Card |
| 5 | 1005 | 106 | Debbie Baker | vvargas@gmail.com | 10/09/2023 | 1051 | Toys | 562.6200000000 | 5 | 2813.1000000000 | Credit Card |
| 6 | 1006 | 71 | Mrs. Tanya Mullen | brendareese@gmail.com | 09/03/2023 | 1069 | UNKNOWN | 4008.6900000000 | 0 | 0.0000000000 | Bank Transfer |
| 7 | 1007 | 188 | Adam Gonzalez | msmith@hebert.org | 13/12/2024 | 1095 | Toys | 2410.5400000000 | 8 | 19284.3200000000 | Credit Card |
| 8 | 1008 | 20 | Anthony Reynolds | cameronwilkins@collins-mcdonald.net | 09/02/2025 | 1089 | UNKNOWN | 4294.9100000000 | 4 | 17179.6400000000 | Credit Card |
| 9 | 1009 | 102 | Ashley Shelton | simsjessica@gmail.com | 31/03/2023 | 1031 | Books | 3815.4300000000 | 9 | 34338.8700000000 | Debit Card |

Query executed successfully. | localhost\SQLEXPRESS (17.0 ... | AMAAN_2005\amaan (86) | uncleaned_dataset | 00:00:00 | Row: 1, Col: 1 | 1,000 rows

*TRY_CONVERT(DATE, purchase_date, 103), bad dates become NULL*

- **TRY_CONVERT** returns NULL instead of an error.
- Style **103** reads dd/mm/yyyy.

**Analyst tip:** use TRY_CONVERT while cleaning, because CONVERT stops on bad data.

# 6 Fix Invalid Emails

A valid email must contain @

SQLQuery9.sq...\amaan (70)) | **SQLQuery8....amaan (54))*** | SQLQuery7.s...\amaan (77))* | HANDLE NEG...amaan (52))*

```sql
--Step 6 Fixing Invalid Email Addresses

select * from sales
where email not like '%@%';

UPDATE sales
SET email= NULL
WHERE email NOT LIKE '%@%'
```

100 % | No issues found | Ln: 8, Ch: 27 | TABS | CRLF | UTF-8

Messages

```
(102 rows affected)

Completion time: 2026-10-25T22:27:21.8049872+05:30
```

100 % | No issues found | Ln: 5, Ch: 1 | TABS | MIXED | UTF-8 with BOM

Query executed successfully. | localhost\SQLEXPRESS (17.0 ... | AMAAN_2005\amaan (54) | uncleaned_dataset | 00:00:00 | Row: 1, Col: 1 | 0 rows

*(102 rows affected)*

- Bad emails set to **NULL**, the row is not deleted.
- The sale is real, only the contact detail is wrong.

# 7 Check & Fix Datatypes

INFORMATION_SCHEMA.COLUMNS

SQLQuery9....amaan (70))* | SQLQuery8.s...\amaan (54))* | SQL(

```
--Step 7 :- Checking the datatype

SELECT COLUMN_NAME, DATA_TYPE
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME='sales';
```

100 % — No issues found

Results | Messages

| | COLUMN_NAME | DATA_TYPE |
|---|---|---|
| 1 | transaction_id | int |
| 2 | customer_id | int |
| 3 | customer_name | nvarchar |
| 4 | email | nvarchar |
| 5 | purchase_date | nvarchar |
| 6 | product_id | int |
| 7 | category | nvarchar |
| 8 | price | decimal |
| 9 | quantity | int |
| 10 | total_amount | decimal |
| 11 | payment_method | nvarchar |
| 12 | delivery_status | nvarchar |
| 13 | customer_addr... | nvarchar |

*purchase_date is nvarchar (text)*

```sql
ALTER TABLE sales
ALTER COLUMN purchase_date DATE;
```

> **Why it matters:** as DATE you can use YEAR(), MONTH(), DATEDIFF() and time-series charts.

# Before vs After

The result of the 7-step cleaning

| Check | Before | After |
|---|---|---|
| Total rows | 1,004 | 1,000 |
| Duplicate transaction_id | 4 | 0 |
| Negative quantity | 318 | 0 |
| NULL category / price | 167 / 58 | 0 / 0 |
| NULL payment / delivery | 138 / 131 | 0 / 0 |
| NULL total_amount | 247 | 0 |
| Payment label variants | 7 | 5 clean |
| Invalid emails | 103 | 0 (set NULL) |
| purchase_date type | nvarchar | DATE |

**Still NULL on purpose:** customer_id (3 rows) and email (102 rows). I did not invent IDs or emails.

# 7 Golden Rules for Data Analysts

What this project taught me

1. **Keep a backup.** Never clean the only copy.

2. **SELECT before UPDATE / DELETE.** Preview rows. Use BEGIN TRAN and ROLLBACK.

3. **Order of steps matters.** Duplicates, then nulls, then calculations.

4. **Choose mean, median or mode with reason.** Outliers? Use median.

5. **Do not delete if you can fix.** Every deleted row is lost information.

6. **Document every assumption.** Cash for missing payment, ABS for negatives.

7. **Validate after cleaning.** Compare row counts, null counts, min and max.

# Skills Used & Next Steps

## SQL skills practised

CTE · ROW_NUMBER() · Dynamic SQL · UPDATE / DELETE · CASE WHEN · TRY_CONVERT · PERCENTILE_CONT · ALTER TABLE · INFORMATION_SCHEMA

## What I will improve next

- Review the 75 rows where quantity = 0.
- Keep text labels consistent (Unknown vs UNKNOWN).
- Investigate the 3 rows with missing customer_id.
- Build a **Power BI dashboard** on this clean data.

### Found this useful?

Like, save and repost. Comment "SQL" for the full script.